7 **(a)** **(i)** Sound cannot be polarised [B1] as it is longitudinal [B1].

(ii) The particles are more tightly packed in solid steel [B1] than in air. Hence it is faster for energy to be transferred between particles.

(iii) I is inversely proportional to square of r. $I \propto \frac{1}{r^2}$ [B1]

iv.1

$$10 \lg \frac{I_X}{\left(1.0 \times 10^{-12}\right)} = 120$$

$I_X = 1.0\ \text{W m}^{-1}$ [B1]

$$20 \lg \frac{p_X}{\left(20 \times 10^{-6}\right)} = 120$$

$p_X = 20\ \text{Pa}$ [B1]

iv.2 Estimated area of an eardrum = 1 cm^2 = 10^{-4} m^2 [B1]

(accepted area between 2 x 10^{-5} m^2 to 2 x 10^{-4} m^2; based on circular area between diameter of 0.5 cm to 1.5 cm)

iv.3 $F = PA$

$= 20\left(10^{-4}\right)$ [C1]

$= 2.0 \times 10^{-3}$ N [A1]

(b) **(i)** To avoid causing noise disturbances to people living on land. [B1]

OR

As no human lives on the ocean, no one will be affected by the noise disturbance. [B1]

(ii) Any of the following (or other sensible suggestions): [B1]

Minimise the drag force at high speed as air is thinner at high altitude.

Minimal obstacle / traffic at high altitude so that it can travel at high speed without obstructions.

Minimise intensity of noise at sea level.

(ii) Speed = (1.4)(295) = 413 m s^{-1} [B1]

(c) **(i)** Resonance [B1]

(ii) Wavelength

= 2(0.91 – 0.54) [C1]

= 0.74 m [A1]

(iii) Speed of sound = fλ

= (480)(0.74) [C1]

= 355 m s^{-1} [A1]

(iv) $v = 331\sqrt{\dfrac{22+273}{273}}$ [M1]

$= 344$ m s^{-1} [A1]